

Modeling Part I

Setup

```
# install.packages(c("survey", "mgcv", "glmnet", "dplyr", "pROC", "ranger", "xgboost"))  
library(survey)
```

Warning: package 'survey' was built under R version 4.5.2

Loading required package: grid

Loading required package: Matrix

Loading required package: survival

Attaching package: 'survey'

The following object is masked from 'package:graphics':

dotchart

```
library(mgcv)
```

Loading required package: nlme

This is mgcv 1.9-3. For overview type 'help("mgcv-package")'.

```
library(glmnet)
```

Loaded glmnet 4.1-10

```
library(dplyr)
```

Attaching package: 'dplyr'

The following object is masked from 'package:nlme':

collapse

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(pROC)
```

Type 'citation("pROC")' for a citation.

Attaching package: 'pROC'

The following objects are masked from 'package:stats':

cov, smooth, var

```
library(ranger)  
library(xgboost)
```

Warning: package 'xgboost' was built under R version 4.5.2

Data split and survey design

```
df <- readRDS("../data/brfss2024_clean.rds")

predictor_formula <- MEDCOST1 ~
  region +
  rurality +
  sex +
  agegroup +
  race +
  income +
  maritalstatus +
  education +
  employment +
  healthstatus +
  physhlth_days +
  menthlth_days +
  depression +
  chronic_count +
  any_disability +
  insurance +
  pers_doc +
  last_checkup

set.seed(263)
train_index <- unlist(lapply(
  split(seq_len(nrow(df)), df$MEDCOST1),
  function(idx) sample(idx, floor(0.8 * length(idx)))
))

train_df <- df[train_index, ]
test_df <- df[-train_index, ]

y_train <- train_df$MEDCOST1
y_test <- test_df$MEDCOST1

split_summary <- data.frame(
  Set = c("Training", "Testing"),
  N = c(nrow(train_df), nrow(test_df)),
  Event_Rate = round(c(mean(y_train), mean(y_test)), 3)
)
```

```
split_summary
```

	Set	N	Event_Rate
1	Training	302236	0.09
2	Testing	75559	0.09

```
# create survey design object for weighted training models
options(survey.lonely.psu = "adjust")

brfss_train_design <- svydesign(
  id = ~PSU,
  strata = ~STSTR,
  weights = ~WEIGHT,
  data = train_df,
  nest = TRUE
)
```

Evaluation helpers

```
safe_divide <- function(num, den) {
  ifelse(den == 0, NA_real_, num / den)
}

get_auc <- function(y, pred_prob) {
  roc_obj <- roc(y, as.numeric(pred_prob), quiet = TRUE)
  as.numeric(auc(roc_obj))
}

get_youden_threshold <- function(y, pred_prob) {
  roc_obj <- roc(y, as.numeric(pred_prob), quiet = TRUE)
  best <- coords(
    roc_obj,
    x = "best",
    best.method = "youden",
    ret = "threshold"
  )
  as.numeric(best[1])
}
```

```

classification_metrics <- function(y, pred_prob, threshold) {
  pred_class <- ifelse(as.numeric(pred_prob) >= threshold, 1, 0)

  tp <- sum(pred_class == 1 & y == 1)
  tn <- sum(pred_class == 0 & y == 0)
  fp <- sum(pred_class == 1 & y == 0)
  fn <- sum(pred_class == 0 & y == 1)

  data.frame(
    Accuracy = mean(pred_class == y),
    Sensitivity = safe_divide(tp, tp + fn),
    Specificity = safe_divide(tn, tn + fp)
  )
}

evaluate_model <- function(model_name, train_y, train_prob, test_y, test_prob) {
  threshold <- get_youden_threshold(train_y, train_prob)
  metrics <- classification_metrics(test_y, test_prob, threshold)

  data.frame(
    Model = model_name,
    Test_AUC = get_auc(test_y, test_prob),
    Threshold = threshold,
    Accuracy = metrics$Accuracy,
    Sensitivity = metrics$Sensitivity,
    Specificity = metrics$Specificity
  )
}

```

Logistic Regression

Weighted

```

weighted_glm <- svyglm(
  predictor_formula,
  design = brfss_train_design,
  family = quasibinomial()
)

summary(weighted_glm)

```

Call:

```
svyglm(formula = predictor_formula, design = brfss_train_design,  
        family = quasibinomial())
```

Survey design:

```
svydesign(id = ~PSU, strata = ~STSTR, weights = ~WEIGHT, data = train_df,  
          nest = TRUE)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-4.6370113	0.0816589	-56.785	< 2e-16	***
regionNortheast	-0.1363063	0.0408467	-3.337	0.000847	***
regionMidwest	-0.2003320	0.0333426	-6.008	1.88e-09	***
regionWest	-0.1553013	0.0439711	-3.532	0.000413	***
ruralityRural	-0.0595700	0.0499737	-1.192	0.233252	
sexFemale	0.2336014	0.0316030	7.392	1.45e-13	***
agegroup18-29	0.4618200	0.0618224	7.470	8.03e-14	***
agegroup30-39	0.5575710	0.0593673	9.392	< 2e-16	***
agegroup40-49	0.5014442	0.0597400	8.394	< 2e-16	***
agegroup50-59	0.1885880	0.0566111	3.331	0.000865	***
agegroup70+	-0.7552759	0.0848425	-8.902	< 2e-16	***
raceBlack_NH	0.3214703	0.0470423	6.834	8.29e-12	***
raceAsian_NH	0.0606820	0.0938916	0.646	0.518087	
raceAIAN_NH	0.0072473	0.1308098	0.055	0.955817	
raceHispanic	0.3139616	0.0446818	7.027	2.12e-12	***
raceOther_NH	0.2847792	0.0662559	4.298	1.72e-05	***
incomeUpper_mid	0.7965609	0.0505694	15.752	< 2e-16	***
incomeMiddle	1.0750588	0.0538232	19.974	< 2e-16	***
incomeLow_mid	1.2320393	0.0709364	17.368	< 2e-16	***
incomeLow	1.0959523	0.0769951	14.234	< 2e-16	***
incomeUnknown	0.7282278	0.0598862	12.160	< 2e-16	***
maritalstatusCurrently_not_married	0.0329601	0.0350691	0.940	0.347289	
educationSome_college	0.1433122	0.0393034	3.646	0.000266	***
educationHS_grad	0.0001488	0.0423403	0.004	0.997196	
educationLess_than_HS	0.0147860	0.0601614	0.246	0.805858	
employmentUnemployed	0.1297953	0.0551859	2.352	0.018675	*
employmentNot_in_labor	-0.2380707	0.0580406	-4.102	4.10e-05	***
employmentRetired	-0.6834323	0.0811420	-8.423	< 2e-16	***
employmentUnable_to_work	-0.4307711	0.0620440	-6.943	3.85e-12	***

healthstatusVery_good	0.3131617	0.0578531	5.413	6.20e-08	***
healthstatusGood	0.5919055	0.0559314	10.583	< 2e-16	***
healthstatusFair	0.9707472	0.0633771	15.317	< 2e-16	***
healthstatusPoor	0.9876553	0.0898556	10.992	< 2e-16	***
physhlth_days	0.0152908	0.0018834	8.119	4.73e-16	***
menthlth_days	0.0273667	0.0016751	16.337	< 2e-16	***
depression	0.3010988	0.0364034	8.271	< 2e-16	***
chronic_count	0.0964308	0.0182342	5.288	1.23e-07	***
any_disability	0.5153845	0.0380575	13.542	< 2e-16	***
insurancePrivate	0.2468420	0.0564338	4.374	1.22e-05	***
insuranceMedicare	-0.2622653	0.0606624	-4.323	1.54e-05	***
insuranceMedicaid_CHIP	-0.2213039	0.0562368	-3.935	8.31e-05	***
insuranceMilitary_VA	-0.7928384	0.1307601	-6.063	1.34e-09	***
insuranceOther_govt	0.2177498	0.0671705	3.242	0.001188	**
insuranceUninsured	1.3722451	0.0522902	26.243	< 2e-16	***
pers_docNo	0.2380285	0.0449675	5.293	1.20e-07	***
last_checkup1_2yrs	0.4702457	0.0434283	10.828	< 2e-16	***
last_checkup2_5yrs	0.6747036	0.0565220	11.937	< 2e-16	***
last_checkupOver_5yrs	0.5696645	0.0622011	9.158	< 2e-16	***
last_checkupNever	0.4700073	0.1773147	2.651	0.008033	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasibinomial family taken to be 0.9492774)

Number of Fisher Scoring iterations: 6

Unweighted

```
unweighted_glm <- glm(
  predictor_formula,
  data = train_df,
  family = binomial()
)

summary(unweighted_glm)
```

Call:

```
glm(formula = predictor_formula, family = binomial(), data = train_df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.7264751	0.0415297	-113.809	< 2e-16	***
regionNortheast	-0.1499946	0.0213110	-7.038	1.95e-12	***
regionMidwest	-0.1811348	0.0194766	-9.300	< 2e-16	***
regionWest	-0.0947175	0.0203774	-4.648	3.35e-06	***
ruralityRural	-0.0577621	0.0220995	-2.614	0.008956	**
sexFemale	0.2243000	0.0153449	14.617	< 2e-16	***
agegroup18-29	0.5897125	0.0292982	20.128	< 2e-16	***
agegroup30-39	0.6697934	0.0278624	24.039	< 2e-16	***
agegroup40-49	0.5468214	0.0271248	20.159	< 2e-16	***
agegroup50-59	0.2668021	0.0263404	10.129	< 2e-16	***
agegroup70+	-0.7425077	0.0340610	-21.799	< 2e-16	***
raceBlack_NH	0.3117026	0.0256580	12.148	< 2e-16	***
raceAsian_NH	0.0248702	0.0476954	0.521	0.602061	
raceAIAN_NH	0.1712614	0.0502463	3.408	0.000653	***
raceHispanic	0.3383085	0.0217818	15.532	< 2e-16	***
raceOther_NH	0.2513350	0.0340068	7.391	1.46e-13	***
incomeUpper_mid	0.7538092	0.0251683	29.951	< 2e-16	***
incomeMiddle	1.1527285	0.0268630	42.911	< 2e-16	***
incomeLow_mid	1.2647825	0.0333850	37.885	< 2e-16	***
incomeLow	1.1278060	0.0377336	29.889	< 2e-16	***
incomeUnknown	0.7502955	0.0304723	24.622	< 2e-16	***
maritalstatusCurrently_not_married	0.0355323	0.0163713	2.170	0.029977	*
educationSome_college	0.0834646	0.0190874	4.373	1.23e-05	***
educationHS_grad	-0.0362193	0.0198820	-1.822	0.068498	.
educationLess_than_HS	0.0338990	0.0294061	1.153	0.248998	
employmentUnemployed	0.1155862	0.0272385	4.243	2.20e-05	***
employmentNot_in_labor	-0.2265222	0.0280272	-8.082	6.36e-16	***
employmentRetired	-0.7064575	0.0325262	-21.720	< 2e-16	***
employmentUnable_to_work	-0.3983088	0.0297627	-13.383	< 2e-16	***
healthstatusVery_good	0.2823574	0.0292240	9.662	< 2e-16	***
healthstatusGood	0.6272805	0.0279389	22.452	< 2e-16	***
healthstatusFair	0.9417902	0.0315466	29.854	< 2e-16	***
healthstatusPoor	1.0068199	0.0413771	24.333	< 2e-16	***
physlth_days	0.0134359	0.0009015	14.904	< 2e-16	***
menthlth_days	0.0266038	0.0007926	33.566	< 2e-16	***
depression	0.2976400	0.0178513	16.673	< 2e-16	***
chronic_count	0.1000302	0.0071867	13.919	< 2e-16	***
any_disability	0.5243505	0.0177140	29.601	< 2e-16	***
insurancePrivate	0.2134449	0.0267084	7.992	1.33e-15	***
insuranceMedicare	-0.3197035	0.0281869	-11.342	< 2e-16	***

insuranceMedicaid_CHIP	-0.3220330	0.0276946	-11.628	< 2e-16	***
insuranceMilitary_VA	-0.9019072	0.0537913	-16.767	< 2e-16	***
insuranceOther_govt	0.1412721	0.0294061	4.804	1.55e-06	***
insuranceUninsured	1.3687649	0.0251772	54.365	< 2e-16	***
pers_docNo	0.2648656	0.0213840	12.386	< 2e-16	***
last_checkup1_2yrs	0.5667479	0.0221378	25.601	< 2e-16	***
last_checkup2_5yrs	0.6820424	0.0272971	24.986	< 2e-16	***
last_checkupOver_5yrs	0.6064199	0.0303609	19.974	< 2e-16	***
last_checkupNever	0.3523421	0.0682454	5.163	2.43e-07	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 183351 on 302235 degrees of freedom
 Residual deviance: 137442 on 302187 degrees of freedom
 AIC: 137540

Number of Fisher Scoring iterations: 6

Comparison

```
# model summary comparison table
weighted_glm_summary <- summary(weighted_glm)$coefficients
unweighted_glm_summary <- summary(unweighted_glm)$coefficients

weighted_glm_df <- data.frame(
  Variable = rownames(weighted_glm_summary),
  Weighted_Estimate = weighted_glm_summary[, 1],
  Weighted_SE = weighted_glm_summary[, 2],
  Weighted_p = weighted_glm_summary[, 4]
)

unweighted_glm_df <- data.frame(
  Variable = rownames(unweighted_glm_summary),
  Unweighted_Estimate = unweighted_glm_summary[, 1],
  Unweighted_SE = unweighted_glm_summary[, 2],
  Unweighted_p = unweighted_glm_summary[, 4]
)

# OR and 95% CI
```

```

weighted_glm_df$Weighted_OR <- exp(weighted_glm_df$Weighted_Estimate)
weighted_ci <- exp(confint(weighted_glm))
weighted_glm_df$Weighted_OR_lower <- weighted_ci[, 1]
weighted_glm_df$Weighted_OR_upper <- weighted_ci[, 2]

unweighted_glm_df$Unweighted_OR <- exp(unweighted_glm_df$Unweighted_Estimate)
unweighted_ci <- exp(confint.default(unweighted_glm))
unweighted_glm_df$Unweighted_OR_lower <- unweighted_ci[, 1]
unweighted_glm_df$Unweighted_OR_upper <- unweighted_ci[, 2]

comparison_glm_table <- merge(
  weighted_glm_df,
  unweighted_glm_df,
  by = "Variable"
)

comparison_glm_table <- comparison_glm_table %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))

comparison_glm_table

```

	Variable	Weighted_Estimate	Weighted_SE	Weighted_p
1	(Intercept)	-4.637	0.082	0.000
2	agegroup18-29	0.462	0.062	0.000
3	agegroup30-39	0.558	0.059	0.000
4	agegroup40-49	0.501	0.060	0.000
5	agegroup50-59	0.189	0.057	0.001
6	agegroup70+	-0.755	0.085	0.000
7	any_disability	0.515	0.038	0.000
8	chronic_count	0.096	0.018	0.000
9	depression	0.301	0.036	0.000
10	educationHS_grad	0.000	0.042	0.997
11	educationLess_than_HS	0.015	0.060	0.806
12	educationSome_college	0.143	0.039	0.000
13	employmentNot_in_labor	-0.238	0.058	0.000
14	employmentRetired	-0.683	0.081	0.000
15	employmentUnable_to_work	-0.431	0.062	0.000
16	employmentUnemployed	0.130	0.055	0.019
17	healthstatusFair	0.971	0.063	0.000
18	healthstatusGood	0.592	0.056	0.000
19	healthstatusPoor	0.988	0.090	0.000
20	healthstatusVery_good	0.313	0.058	0.000

21	incomeLow	1.096	0.077	0.000
22	incomeLow_mid	1.232	0.071	0.000
23	incomeMiddle	1.075	0.054	0.000
24	incomeUnknown	0.728	0.060	0.000
25	incomeUpper_mid	0.797	0.051	0.000
26	insuranceMedicaid_CHIP	-0.221	0.056	0.000
27	insuranceMedicare	-0.262	0.061	0.000
28	insuranceMilitary_VA	-0.793	0.131	0.000
29	insuranceOther_govt	0.218	0.067	0.001
30	insurancePrivate	0.247	0.056	0.000
31	insuranceUninsured	1.372	0.052	0.000
32	last_checkup1_2yrs	0.470	0.043	0.000
33	last_checkup2_5yrs	0.675	0.057	0.000
34	last_checkupNever	0.470	0.177	0.008
35	last_checkupOver_5yrs	0.570	0.062	0.000
36	maritalstatusCurrently_not_married	0.033	0.035	0.347
37	menthlth_days	0.027	0.002	0.000
38	pers_docNo	0.238	0.045	0.000
39	physhlth_days	0.015	0.002	0.000
40	raceAIAN_NH	0.007	0.131	0.956
41	raceAsian_NH	0.061	0.094	0.518
42	raceBlack_NH	0.321	0.047	0.000
43	raceHispanic	0.314	0.045	0.000
44	raceOther_NH	0.285	0.066	0.000
45	regionMidwest	-0.200	0.033	0.000
46	regionNortheast	-0.136	0.041	0.001
47	regionWest	-0.155	0.044	0.000
48	ruralityRural	-0.060	0.050	0.233
49	sexFemale	0.234	0.032	0.000
	Weighted_OR	Weighted_OR_lower	Weighted_OR_upper	Unweighted_Estimate
1	0.010	0.008	0.011	-4.726
2	1.587	1.406	1.791	0.590
3	1.746	1.555	1.962	0.670
4	1.651	1.469	1.856	0.547
5	1.208	1.081	1.349	0.267
6	0.470	0.398	0.555	-0.743
7	1.674	1.554	1.804	0.524
8	1.101	1.063	1.141	0.100
9	1.351	1.258	1.451	0.298
10	1.000	0.921	1.087	-0.036
11	1.015	0.902	1.142	0.034
12	1.154	1.069	1.247	0.083
13	0.788	0.703	0.883	-0.227

14	0.505	0.431	0.592	-0.706
15	0.650	0.576	0.734	-0.398
16	1.139	1.022	1.269	0.116
17	2.640	2.332	2.989	0.942
18	1.807	1.620	2.017	0.627
19	2.685	2.251	3.202	1.007
20	1.368	1.221	1.532	0.282
21	2.992	2.573	3.479	1.128
22	3.428	2.983	3.940	1.265
23	2.930	2.637	3.256	1.153
24	2.071	1.842	2.329	0.750
25	2.218	2.009	2.449	0.754
26	0.801	0.718	0.895	-0.322
27	0.769	0.683	0.866	-0.320
28	0.453	0.350	0.585	-0.902
29	1.243	1.090	1.418	0.141
30	1.280	1.146	1.430	0.213
31	3.944	3.560	4.370	1.369
32	1.600	1.470	1.743	0.567
33	1.963	1.758	2.193	0.682
34	1.600	1.130	2.265	0.352
35	1.768	1.565	1.997	0.606
36	1.034	0.965	1.107	0.036
37	1.028	1.024	1.031	0.027
38	1.269	1.162	1.386	0.265
39	1.015	1.012	1.019	0.013
40	1.007	0.779	1.302	0.171
41	1.063	0.884	1.277	0.025
42	1.379	1.258	1.512	0.312
43	1.369	1.254	1.494	0.338
44	1.329	1.168	1.514	0.251
45	0.818	0.767	0.874	-0.181
46	0.873	0.805	0.945	-0.150
47	0.856	0.785	0.933	-0.095
48	0.942	0.854	1.039	-0.058
49	1.263	1.187	1.344	0.224
Unweighted_SE Unweighted_p Unweighted_OR Unweighted_OR_lower				
1	0.042	0.000	0.009	0.008
2	0.029	0.000	1.803	1.703
3	0.028	0.000	1.954	1.850
4	0.027	0.000	1.728	1.638
5	0.026	0.000	1.306	1.240
6	0.034	0.000	0.476	0.445

7	0.018	0.000	1.689	1.632
8	0.007	0.000	1.105	1.090
9	0.018	0.000	1.347	1.300
10	0.020	0.068	0.964	0.928
11	0.029	0.249	1.034	0.977
12	0.019	0.000	1.087	1.047
13	0.028	0.000	0.797	0.755
14	0.033	0.000	0.493	0.463
15	0.030	0.000	0.671	0.633
16	0.027	0.000	1.123	1.064
17	0.032	0.000	2.565	2.411
18	0.028	0.000	1.873	1.773
19	0.041	0.000	2.737	2.524
20	0.029	0.000	1.326	1.252
21	0.038	0.000	3.089	2.869
22	0.033	0.000	3.542	3.318
23	0.027	0.000	3.167	3.004
24	0.030	0.000	2.118	1.995
25	0.025	0.000	2.125	2.023
26	0.028	0.000	0.725	0.686
27	0.028	0.000	0.726	0.687
28	0.054	0.000	0.406	0.365
29	0.029	0.000	1.152	1.087
30	0.027	0.000	1.238	1.175
31	0.025	0.000	3.930	3.741
32	0.022	0.000	1.763	1.688
33	0.027	0.000	1.978	1.875
34	0.068	0.000	1.422	1.244
35	0.030	0.000	1.834	1.728
36	0.016	0.030	1.036	1.003
37	0.001	0.000	1.027	1.025
38	0.021	0.000	1.303	1.250
39	0.001	0.000	1.014	1.012
40	0.050	0.001	1.187	1.075
41	0.048	0.602	1.025	0.934
42	0.026	0.000	1.366	1.299
43	0.022	0.000	1.403	1.344
44	0.034	0.000	1.286	1.203
45	0.019	0.000	0.834	0.803
46	0.021	0.000	0.861	0.826
47	0.020	0.000	0.910	0.874
48	0.022	0.009	0.944	0.904
49	0.015	0.000	1.251	1.214

	Unweighted_OR_upper
1	0.010
2	1.910
3	2.063
4	1.822
5	1.375
6	0.509
7	1.749
8	1.121
9	1.395
10	1.003
11	1.096
12	1.128
13	0.842
14	0.526
15	0.712
16	1.184
17	2.728
18	1.978
19	2.968
20	1.404
21	3.326
22	3.782
23	3.338
24	2.248
25	2.233
26	0.765
27	0.768
28	0.451
29	1.220
30	1.304
31	4.129
32	1.841
33	2.087
34	1.626
35	1.946
36	1.070
37	1.029
38	1.359
39	1.015
40	1.310
41	1.126
42	1.436

43	1.464
44	1.374
45	0.867
46	0.897
47	0.947
48	0.986
49	1.290

```
pred_train_w_glm <- as.numeric(predict(weighted_glm, type = "response"))
pred_test_w_glm <- as.numeric(predict(
  weighted_glm,
  newdata = test_df,
  type = "response"
))

pred_train_u_glm <- as.numeric(predict(unweighted_glm, type = "response"))
pred_test_u_glm <- as.numeric(predict(
  unweighted_glm,
  newdata = test_df,
  type = "response"
))
```

Logistic Regression with Lasso

```
x_train <- model.matrix(predictor_formula, data = train_df)[, -1]
x_test <- model.matrix(predictor_formula, data = test_df)[, -1]
```

Weighted

```
set.seed(263)
weighted_lasso <- cv.glmnet(
  x_train,
  y_train,
  family = "binomial",
  alpha = 1,
  nfolds = 5,
  type.measure = "auc",
```

```
weights = train_df$WEIGHT
)
```

Unweighted

```
set.seed(263)
unweighted_lasso <- cv.glmnet(
  x_train,
  y_train,
  family = "binomial",
  alpha = 1,
  nfolds = 5,
  type.measure = "auc"
)
```

Comparison

```
# Lasso selected variables
lasso_out <- cbind(
  coef(weighted_lasso, s = "lambda.1se"),
  coef(unweighted_lasso, s = "lambda.1se")
)
colnames(lasso_out) <- c("Weighted", "Unweighted")
lasso_out
```

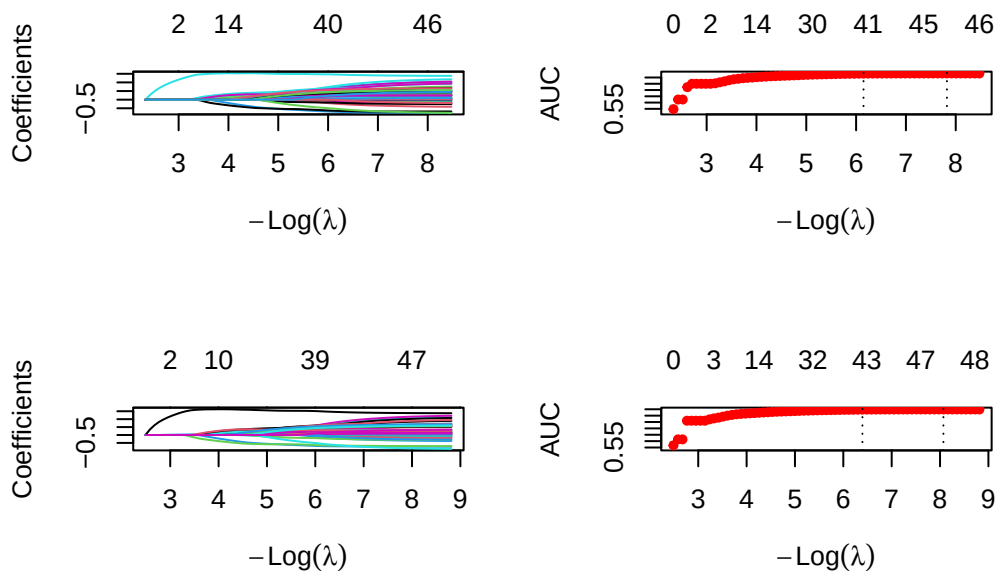
```
49 x 2 sparse Matrix of class "dgCMatrix"

              Weighted  Unweighted
(Intercept) -3.84960928 -4.02266552
regionNortheast -0.01361792 -0.02700524
regionMidwest -0.07221179 -0.04648782
regionWest -0.02822917 .
ruralityRural . .
sexFemale 0.17601392 0.17113155
agegroup18-29 0.19092644 0.34854318
agegroup30-39 0.27954769 0.41991106
agegroup40-49 0.22557800 0.30240362
agegroup50-59 . 0.04789947
agegroup70+ -0.70181906 -0.71281730
```


raceBlack_NH	0.24986261	0.26303677
raceAsian_NH	.	.
raceAIAN_NH	.	0.02699749
raceHispanic	0.27975033	0.32761997
raceOther_NH	0.16373612	0.15693613
incomeUpper_mid	0.45134992	0.38931161
incomeMiddle	0.70516453	0.74607267
incomeLow_mid	0.79282341	0.78813728
incomeLow	0.63401702	0.62264035
incomeUnknown	0.30660651	0.29625374
maritalstatusCurrently_not_married	0.06773648	0.07215828
educationSome_college	0.09074793	0.05945317
educationHS_grad	.	.
educationLess_than_HS	.	0.02297150
employmentUnemployed	0.15564771	0.15564405
employmentNot_in_labor	-0.07653310	-0.05857963
employmentRetired	-0.64363840	-0.66661076
employmentUnable_to_work	-0.23369290	-0.20193423
healthstatusVery_good	.	.
healthstatusGood	0.28456161	0.33125856
healthstatusFair	0.65620981	0.63931822
healthstatusPoor	0.61989177	0.65786712
physhlth_days	0.01357281	0.01224454
menthlth_days	0.02890296	0.02816446
depression	0.28743583	0.28999731
chronic_count	0.04697092	0.06122349
any_disability	0.48809300	0.49333760
insurancePrivate	0.19375060	0.16539052
insuranceMedicare	-0.17649645	-0.24163014
insuranceMedicaid_CHIP	-0.01123660	-0.10490924
insuranceMilitary_VA	-0.52751203	-0.62675740
insuranceOther_govt	0.25200232	0.19334632
insuranceUninsured	1.45892628	1.46012326
pers_docNo	0.26273983	0.28898379
last_checkup1_2yrs	0.38169559	0.47969162
last_checkup2_5yrs	0.56958270	0.58453357
last_checkupOver_5yrs	0.44492947	0.48686085
last_checkupNever	0.25741087	0.17798610

```
# Lasso plot
par(mfrow = c(2, 2))
plot(weighted_lasso$glmnet.fit, xvar = "lambda")
```

```
plot(weighted_lasso)
plot(unweighted_lasso$glmnet.fit, xvar = "lambda")
plot(unweighted_lasso)
```



```
pred_train_w_lasso <- as.numeric(predict(
  weighted_lasso,
  newx = x_train,
  s = "lambda.1se",
  type = "response"
))
pred_test_w_lasso <- as.numeric(predict(
  weighted_lasso,
  newx = x_test,
  s = "lambda.1se",
  type = "response"
))

pred_train_u_lasso <- as.numeric(predict(
  unweighted_lasso,
  newx = x_train,
  s = "lambda.1se",
  type = "response"
))
pred_test_u_lasso <- as.numeric(predict(
  unweighted_lasso,
```

```

newx = x_test,
s = "lambda.1se",
type = "response"
))

```

GAM

Weighted

```

weighted_gam <- gam(
  MEDCOST1 ~
    s(physhlth_days) +
    s(menthlth_days) +
    region +
    rurality +
    sex +
    agegroup +
    race +
    income +
    maritalstatus +
    education +
    employment +
    healthstatus +
    depression +
    chronic_count +
    any_disability +
    insurance +
    pers_doc +
    last_checkup,
  data = train_df,
  family = binomial(),
  weights = train_df$WEIGHT
)

```

```

Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!
Warning in eval(family$initialize): non-integer #successes in a binomial glm!

```

```
summary(weighted_gam)
```

Family: binomial
Link function: logit

Formula:

```
MEDCOST1 ~ s(physhlth_days) + s(menthlth_days) + region + rurality +  
sex + agegroup + race + income + maritalstatus + education +  
employment + healthstatus + depression + chronic_count +  
any_disability + insurance + pers_doc + last_checkup
```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.3708206	0.0015900	-2748.88	<2e-16	***
regionNortheast	-0.1454134	0.0008280	-175.61	<2e-16	***
regionMidwest	-0.2026572	0.0007429	-272.78	<2e-16	***
regionWest	-0.1646163	0.0007101	-231.81	<2e-16	***
ruralityRural	-0.0542352	0.0011434	-47.43	<2e-16	***
sexFemale	0.2098040	0.0005742	365.38	<2e-16	***
agegroup18-29	0.3703938	0.0011465	323.05	<2e-16	***
agegroup30-39	0.5056518	0.0011097	455.67	<2e-16	***
agegroup40-49	0.4744211	0.0010971	432.43	<2e-16	***
agegroup50-59	0.1844345	0.0010957	168.33	<2e-16	***
agegroup70+	-0.7401930	0.0015840	-467.29	<2e-16	***
raceBlack_NH	0.3257913	0.0008580	379.71	<2e-16	***
raceAsian_NH	0.0755378	0.0013285	56.86	<2e-16	***
raceAIAN_NH	0.0212827	0.0023158	9.19	<2e-16	***
raceHispanic	0.3263093	0.0007352	443.82	<2e-16	***
raceOther_NH	0.3006757	0.0012377	242.93	<2e-16	***
incomeUpper_mid	0.7971882	0.0009277	859.33	<2e-16	***
incomeMiddle	1.0858987	0.0009879	1099.22	<2e-16	***
incomeLow_mid	1.2583941	0.0012238	1028.29	<2e-16	***
incomeLow	1.1193561	0.0013781	812.23	<2e-16	***
incomeUnknown	0.7603928	0.0010949	694.50	<2e-16	***
maritalstatusCurrently_not_married	0.0171194	0.0006161	27.79	<2e-16	***
educationSome_college	0.1592516	0.0007687	207.17	<2e-16	***
educationHS_grad	0.0373577	0.0008110	46.06	<2e-16	***
educationLess_than_HS	0.0586191	0.0010251	57.19	<2e-16	***
employmentUnemployed	0.1243212	0.0009597	129.54	<2e-16	***
employmentNot_in_labor	-0.2590539	0.0009265	-279.62	<2e-16	***
employmentRetired	-0.6612887	0.0014156	-467.13	<2e-16	***

employmentUnable_to_work	-0.4204178	0.0011407	-368.57	<2e-16 ***
healthstatusVery_good	0.2487961	0.0010472	237.59	<2e-16 ***
healthstatusGood	0.4954591	0.0009966	497.16	<2e-16 ***
healthstatusFair	0.8241300	0.0011273	731.05	<2e-16 ***
healthstatusPoor	0.9537749	0.0015182	628.22	<2e-16 ***
depression	0.2275225	0.0006830	333.11	<2e-16 ***
chronic_count	0.0862863	0.0002912	296.30	<2e-16 ***
any_disability	0.4784376	0.0006542	731.36	<2e-16 ***
insurancePrivate	0.2482076	0.0009650	257.20	<2e-16 ***
insuranceMedicare	-0.2476221	0.0011130	-222.48	<2e-16 ***
insuranceMedicaid_CHIP	-0.2004861	0.0010372	-193.30	<2e-16 ***
insuranceMilitary_VA	-0.7733336	0.0018528	-417.39	<2e-16 ***
insuranceOther_govt	0.2318134	0.0010486	221.07	<2e-16 ***
insuranceUninsured	1.3992627	0.0009040	1547.78	<2e-16 ***
pers_docNo	0.2636427	0.0007517	350.72	<2e-16 ***
last_checkup1_2yrs	0.4706593	0.0008150	577.50	<2e-16 ***
last_checkup2_5yrs	0.6698035	0.0009670	692.68	<2e-16 ***
last_checkupOver_5yrs	0.5740601	0.0011048	519.59	<2e-16 ***
last_checkupNever	0.5263804	0.0023726	221.86	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(physhlth_days)	8.747	8.973	548819	<2e-16 ***
s(menthlth_days)	8.824	8.987	1040177	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.219 Deviance explained = 24.2%

UBRE = 313.85 Scale est. = 1 n = 302236

Unweighted

```
unweighted_gam <- gam(
  MEDCOST1 ~
    s(physhlth_days) +
    s(menthlth_days) +
    region +
    rurality +
    sex +
```

```

    agegroup +
    race +
    income +
    maritalstatus +
    education +
    employment +
    healthstatus +
    depression +
    chronic_count +
    any_disability +
    insurance +
    pers_doc +
    last_checkup,
  data = train_df,
  family = binomial()
)

summary(unweighted_gam)

```

Family: binomial
Link function: logit

Formula:

```

MEDCOST1 ~ s(physhlth_days) + s(menthlth_days) + region + rurality +
  sex + agegroup + race + income + maritalstatus + education +
  employment + healthstatus + depression + chronic_count +
  any_disability + insurance + pers_doc + last_checkup

```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.449756	0.041995	-105.958	< 2e-16	***
regionNortheast	-0.157795	0.021362	-7.387	1.51e-13	***
regionMidwest	-0.180624	0.019532	-9.247	< 2e-16	***
regionWest	-0.106331	0.020433	-5.204	1.95e-07	***
ruralityRural	-0.046363	0.022151	-2.093	0.036350	*
sexFemale	0.193387	0.015413	12.547	< 2e-16	***
agegroup18-29	0.477097	0.029660	16.086	< 2e-16	***
agegroup30-39	0.595038	0.028071	21.198	< 2e-16	***
agegroup40-49	0.502310	0.027249	18.434	< 2e-16	***
agegroup50-59	0.245778	0.026419	9.303	< 2e-16	***
agegroup70+	-0.719457	0.034115	-21.089	< 2e-16	***

raceBlack_NH	0.320840	0.025723	12.473	< 2e-16	***
raceAsian_NH	0.052196	0.047885	1.090	0.275699	
raceAIAN_NH	0.180532	0.050321	3.588	0.000334	***
raceHispanic	0.366228	0.021852	16.759	< 2e-16	***
raceOther_NH	0.262439	0.034034	7.711	1.25e-14	***
incomeUpper_mid	0.751290	0.025221	29.789	< 2e-16	***
incomeMiddle	1.156102	0.026931	42.929	< 2e-16	***
incomeLow_mid	1.271483	0.033463	37.997	< 2e-16	***
incomeLow	1.143685	0.037785	30.268	< 2e-16	***
incomeUnknown	0.775775	0.030549	25.395	< 2e-16	***
maritalstatusCurrently_not_married	0.021287	0.016436	1.295	0.195262	
educationSome_college	0.103646	0.019152	5.412	6.24e-08	***
educationHS_grad	0.003871	0.019993	0.194	0.846481	
educationLess_than_HS	0.084469	0.029582	2.855	0.004298	**
employmentUnemployed	0.110712	0.027322	4.052	5.08e-05	***
employmentNot_in_labor	-0.238979	0.028089	-8.508	< 2e-16	***
employmentRetired	-0.693713	0.032613	-21.271	< 2e-16	***
employmentUnable_to_work	-0.380120	0.029717	-12.791	< 2e-16	***
healthstatusVery_good	0.209140	0.029448	7.102	1.23e-12	***
healthstatusGood	0.514293	0.028309	18.167	< 2e-16	***
healthstatusFair	0.787034	0.032037	24.566	< 2e-16	***
healthstatusPoor	0.953753	0.041396	23.040	< 2e-16	***
depression	0.220052	0.017947	12.261	< 2e-16	***
chronic_count	0.088391	0.007203	12.271	< 2e-16	***
any_disability	0.490861	0.017743	27.665	< 2e-16	***
insurancePrivate	0.219553	0.026794	8.194	2.52e-16	***
insuranceMedicare	-0.305154	0.028250	-10.802	< 2e-16	***
insuranceMedicaid_CHIP	-0.305355	0.027707	-11.021	< 2e-16	***
insuranceMilitary_VA	-0.875437	0.053694	-16.304	< 2e-16	***
insuranceOther_govt	0.152086	0.029465	5.162	2.45e-07	***
insuranceUninsured	1.404363	0.025370	55.355	< 2e-16	***
pers_docNo	0.287362	0.021479	13.379	< 2e-16	***
last_checkup1_2yrs	0.563799	0.022203	25.393	< 2e-16	***
last_checkup2_5yrs	0.676201	0.027408	24.672	< 2e-16	***
last_checkupOver_5yrs	0.612276	0.030514	20.066	< 2e-16	***
last_checkupNever	0.396017	0.068746	5.761	8.38e-09	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(physhlth_days)	6.641	7.506	635.2	<2e-16 ***
s(menthlth_days)	8.285	8.801	1453.8	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.21 Deviance explained = 25.6%

UBRE = -0.54815 Scale est. = 1 n = 302236

Comparison

```
# coefficients comparison table
weighted_gam_coef <- summary(weighted_gam)$p.coef
unweighted_gam_coef <- summary(unweighted_gam)$p.coef

weighted_gam_df <- data.frame(
  Variable = names(weighted_gam_coef),
  Weighted_Est = weighted_gam_coef,
  Weighted_OR = exp(weighted_gam_coef)
)

unweighted_gam_df <- data.frame(
  Variable = names(unweighted_gam_coef),
  Unweighted_Est = unweighted_gam_coef,
  Unweighted_OR = exp(unweighted_gam_coef)
)

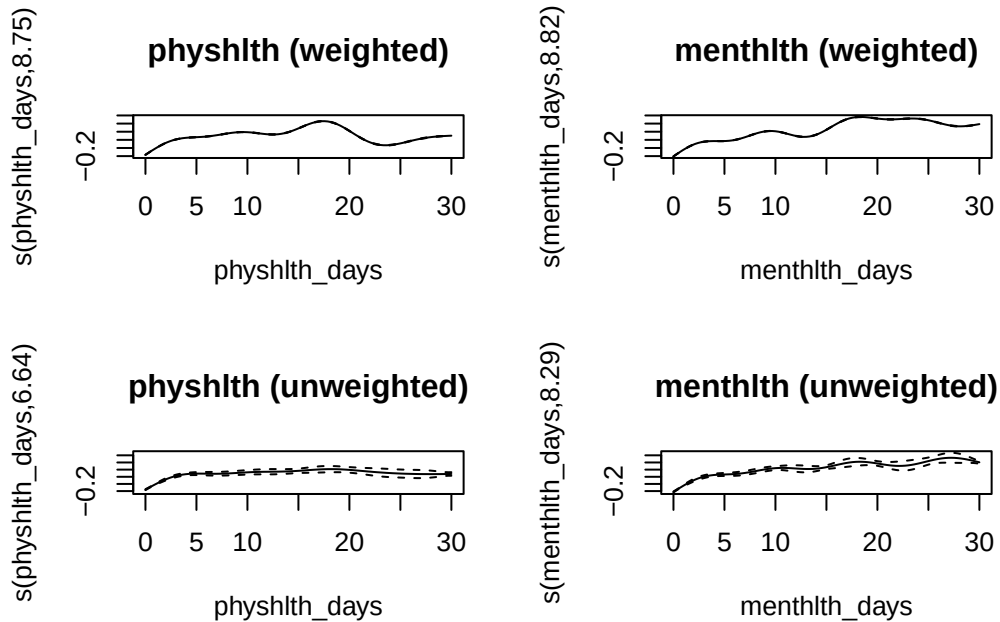
gam_comparison <- merge(weighted_gam_df, unweighted_gam_df, by = "Variable")
gam_comparison <- gam_comparison %>%
  mutate(across(where(is.numeric), ~ round(.x, 3)))
gam_comparison
```

	Variable	Weighted_Est	Weighted_OR	Unweighted_Est
1	(Intercept)	-4.371	0.013	-4.450
2	agegroup18-29	0.370	1.448	0.477
3	agegroup30-39	0.506	1.658	0.595
4	agegroup40-49	0.474	1.607	0.502
5	agegroup50-59	0.184	1.203	0.246
6	agegroup70+	-0.740	0.477	-0.719
7	any_disability	0.478	1.614	0.491
8	chronic_count	0.086	1.090	0.088
9	depression	0.228	1.255	0.220
10	educationHS_grad	0.037	1.038	0.004

11	educationLess_than_HS	0.059	1.060	0.084
12	educationSome_college	0.159	1.173	0.104
13	employmentNot_in_labor	-0.259	0.772	-0.239
14	employmentRetired	-0.661	0.516	-0.694
15	employmentUnable_to_work	-0.420	0.657	-0.380
16	employmentUnemployed	0.124	1.132	0.111
17	healthstatusFair	0.824	2.280	0.787
18	healthstatusGood	0.495	1.641	0.514
19	healthstatusPoor	0.954	2.595	0.954
20	healthstatusVery_good	0.249	1.282	0.209
21	incomeLow	1.119	3.063	1.144
22	incomeLow_mid	1.258	3.520	1.271
23	incomeMiddle	1.086	2.962	1.156
24	incomeUnknown	0.760	2.139	0.776
25	incomeUpper_mid	0.797	2.219	0.751
26	insuranceMedicaid_CHIP	-0.200	0.818	-0.305
27	insuranceMedicare	-0.248	0.781	-0.305
28	insuranceMilitary_VA	-0.773	0.461	-0.875
29	insuranceOther_govt	0.232	1.261	0.152
30	insurancePrivate	0.248	1.282	0.220
31	insuranceUninsured	1.399	4.052	1.404
32	last_checkup1_2yrs	0.471	1.601	0.564
33	last_checkup2_5yrs	0.670	1.954	0.676
34	last_checkupNever	0.526	1.693	0.396
35	last_checkupOver_5yrs	0.574	1.775	0.612
36	maritalstatusCurrently_not_married	0.017	1.017	0.021
37	pers_docNo	0.264	1.302	0.287
38	raceAIAN_NH	0.021	1.022	0.181
39	raceAsian_NH	0.076	1.078	0.052
40	raceBlack_NH	0.326	1.385	0.321
41	raceHispanic	0.326	1.386	0.366
42	raceOther_NH	0.301	1.351	0.262
43	regionMidwest	-0.203	0.817	-0.181
44	regionNortheast	-0.145	0.865	-0.158
45	regionWest	-0.165	0.848	-0.106
46	ruralityRural	-0.054	0.947	-0.046
47	sexFemale	0.210	1.233	0.193
Unweighted_OR				
1	0.012			
2	1.611			
3	1.813			
4	1.653			
5	1.279			

6	0.487
7	1.634
8	1.092
9	1.246
10	1.004
11	1.088
12	1.109
13	0.787
14	0.500
15	0.684
16	1.117
17	2.197
18	1.672
19	2.595
20	1.233
21	3.138
22	3.566
23	3.178
24	2.172
25	2.120
26	0.737
27	0.737
28	0.417
29	1.164
30	1.246
31	4.073
32	1.757
33	1.966
34	1.486
35	1.845
36	1.022
37	1.333
38	1.198
39	1.054
40	1.378
41	1.442
42	1.300
43	0.835
44	0.854
45	0.899
46	0.955
47	1.213

```
# smooth term plot
par(mfrow = c(2, 2))
plot(weighted_gam, select = 1, main = "physhlth (weighted)")
plot(weighted_gam, select = 2, main = "menthlth (weighted)")
plot(unweighted_gam, select = 1, main = "physhlth (unweighted)")
plot(unweighted_gam, select = 2, main = "menthlth (unweighted)")
```



```
# p-values for smooth terms
summary(weighted_gam)$s.pv
```

```
[1] 0 0
```

```
summary(unweighted_gam)$s.pv
```

```
[1] 0 0
```

```
pred_train_w_gam <- as.numeric(predict(weighted_gam, type = "response"))
pred_test_w_gam <- as.numeric(predict(
  weighted_gam,
  newdata = test_df,
  type = "response"
))
```

```

pred_train_u_gam <- as.numeric(predict(unweighted_gam, type = "response"))
pred_test_u_gam <- as.numeric(predict(
  unweighted_gam,
  newdata = test_df,
  type = "response"
))

```

Random Forest

```

rf_train_df <- train_df %>%
  mutate(MEDCOST1 = factor(MEDCOST1, levels = c(0, 1)))

rf_test_df <- test_df %>%
  mutate(MEDCOST1 = factor(MEDCOST1, levels = c(0, 1)))

rf_grid <- expand.grid(
  mtry = c(4, 6, 9),
  min.node.size = c(50, 100),
  KEEP.OUT.ATTRS = FALSE
)

set.seed(263)
rf_tuning <- lapply(seq_len(nrow(rf_grid)), function(i) {
  fit <- ranger(
    predictor_formula,
    data = rf_train_df,
    probability = TRUE,
    num.trees = 300,
    mtry = rf_grid$mtry[i],
    min.node.size = rf_grid$min.node.size[i],
    respect.unordered.factors = "order",
    seed = 263
  )

  data.frame(
    mtry = rf_grid$mtry[i],
    min.node.size = rf_grid$min.node.size[i],
    OOB_Error = fit$prediction.error
  )
})

```

```

}) %>%
  bind_rows() %>%
  arrange(OOB_Error)

```

```

Growing trees.. Progress: 69%. Estimated remaining time: 13 seconds.
Growing trees.. Progress: 54%. Estimated remaining time: 26 seconds.
Growing trees.. Progress: 41%. Estimated remaining time: 45 seconds.
Growing trees.. Progress: 81%. Estimated remaining time: 14 seconds.
Growing trees.. Progress: 74%. Estimated remaining time: 11 seconds.
Growing trees.. Progress: 57%. Estimated remaining time: 23 seconds.
Growing trees.. Progress: 43%. Estimated remaining time: 41 seconds.
Growing trees.. Progress: 86%. Estimated remaining time: 10 seconds.

```

```
rf_tuning
```

	mtry	min.node.size	OOB_Error
1	4	100	0.06440932
2	6	100	0.06441033
3	4	50	0.06447070
4	6	50	0.06454268
5	9	100	0.06454389
6	9	50	0.06474709

```

best_rf <- rf_tuning[1, ]

set.seed(263)
random_forest <- ranger(
  predictor_formula,
  data = rf_train_df,
  probability = TRUE,
  num.trees = 500,
  mtry = best_rf$mtry,
  min.node.size = best_rf$min.node.size,
  importance = "impurity",
  respect.unordered.factors = "order",
  seed = 263
)

```

```

Growing trees.. Progress: 43%. Estimated remaining time: 41 seconds.
Growing trees.. Progress: 86%. Estimated remaining time: 10 seconds.

```

```

pred_train_rf <- predict(random_forest, data = rf_train_df)$predictions[, "1"]
pred_test_rf <- predict(random_forest, data = rf_test_df)$predictions[, "1"]

rf_importance <- data.frame(
  Variable = names(random_forest$variable.importance),
  Importance = as.numeric(random_forest$variable.importance)
) %>%
  arrange(desc(Importance)) %>%
  mutate(Importance = round(Importance, 3))

head(rf_importance, 15)

```

	Variable	Importance
1	insurance	3451.676
2	menthlth_days	1785.735
3	agegroup	1030.043
4	physhlth_days	976.965
5	income	968.388
6	healthstatus	957.686
7	last_checkup	807.454
8	employment	732.712
9	pers_doc	614.652
10	any_disability	534.360
11	race	457.123
12	chronic_count	433.469
13	depression	364.587
14	education	325.380
15	region	292.720

Boosting

```

dtrain <- xgb.DMatrix(data = x_train, label = y_train)
dtest <- xgb.DMatrix(data = x_test, label = y_test)

xgb_grid <- expand.grid(
  eta = c(0.05, 0.10),
  max_depth = c(2, 3),
  KEEP.OUT.ATTRS = FALSE
)

```

```

scale_pos_weight <- sum(y_train == 0) / sum(y_train == 1)

set.seed(263)
xgb_tuning <- lapply(seq_len(nrow(xgb_grid)), function(i) {
  params <- list(
    objective = "binary:logistic",
    eval_metric = "auc",
    eta = xgb_grid$eta[i],
    max_depth = xgb_grid$max_depth[i],
    min_child_weight = 5,
    subsample = 0.8,
    colsample_bytree = 0.8,
    scale_pos_weight = scale_pos_weight
  )

  cv_fit <- xgb.cv(
    params = params,
    data = dtrain,
    nrounds = 200,
    nfold = 3,
    early_stopping_rounds = 10,
    verbose = 0
  )

  cv_results <- cv_fit$evaluation_log
  best_row <- which.max(cv_results$test_auc_mean)

  data.frame(
    eta = xgb_grid$eta[i],
    max_depth = xgb_grid$max_depth[i],
    best_iteration = cv_results$iter[best_row],
    CV_AUC = cv_results$test_auc_mean[best_row]
  )
}) %>%
  bind_rows() %>%
  arrange(desc(CV_AUC))

xgb_tuning

```

	eta	max_depth	best_iteration	CV_AUC
1	0.10	3	200	0.8539367

2	0.10	2	200	0.8497087
3	0.05	3	200	0.8495945
4	0.05	2	200	0.8433611

```
best_xgb <- xgb_tuning[1, ]

xgb_params <- list(
  objective = "binary:logistic",
  eval_metric = "auc",
  eta = best_xgb$eta,
  max_depth = as.integer(best_xgb$max_depth),
  min_child_weight = 5,
  subsample = 0.8,
  colsample_bytree = 0.8,
  scale_pos_weight = scale_pos_weight
)

set.seed(263)
xgb_model <- xgb.train(
  params = xgb_params,
  data = dtrain,
  nrounds = as.integer(best_xgb$best_iteration),
  verbose = 0
)

pred_train_xgb <- predict(xgb_model, dtrain)
pred_test_xgb <- predict(xgb_model, dtest)

xgb_importance <- xgb.importance(
  feature_names = colnames(x_train),
  model = xgb_model
) %>%
  as.data.frame()

head(xgb_importance, 15)
```

	Feature	Gain	Cover	Frequency
1	menthlth_days	0.19621177	0.04401531	0.05785714
2	insuranceUninsured	0.19304188	0.04073688	0.02500000
3	employmentRetired	0.11836181	0.02240954	0.02428571
4	agegroup70+	0.06628992	0.02898012	0.03714286
5	any_disability	0.05620225	0.02577227	0.02428571

6	physhlth_days	0.04923127	0.02947217	0.05142857
7	pers_docNo	0.03944139	0.01663315	0.02714286
8	raceHispanic	0.03051371	0.01811074	0.01357143
9	incomeMiddle	0.03004036	0.06244526	0.04000000
10	incomeLow_mid	0.02247781	0.04876644	0.03357143
11	maritalstatusCurrently_not_married	0.01733071	0.01613894	0.03214286
12	healthstatusFair	0.01562171	0.02198432	0.01714286
13	depression	0.01290773	0.01657696	0.02142857
14	incomeUpper_mid	0.01174884	0.04549752	0.04071429
15	incomeLow	0.01096394	0.03491154	0.02857143

Tree-based model figures

```

top_rf <- head(rf_importance, 10)
top_xgb <- head(xgb_importance, 10)

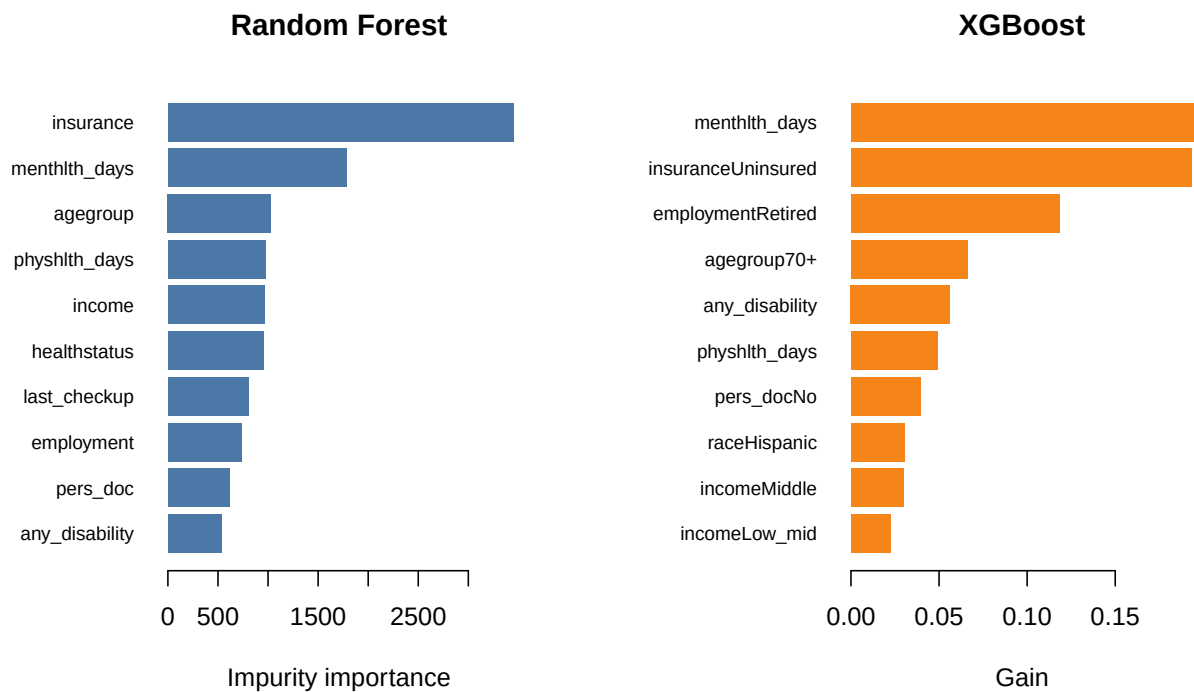
old_par <- par(no.readonly = TRUE)
par(mfrow = c(1, 2), mar = c(5, 10, 4, 1))

barplot(
  rev(top_rf$Importance),
  names.arg = rev(top_rf$Variable),
  horiz = TRUE,
  las = 1,
  cex.names = 0.75,
  main = "Random Forest",
  xlab = "Impurity importance",
  col = "#4C78A8",
  border = NA
)

barplot(
  rev(top_xgb$Gain),
  names.arg = rev(top_xgb$Feature),
  horiz = TRUE,
  las = 1,
  cex.names = 0.75,
  main = "XGBoost",
  xlab = "Gain",
  col = "#F58518",

```

```
border = NA
)
```



```
par(old_par)
```

```
roc_rf <- roc(y_test, pred_test_rf, quiet = TRUE)
roc_xgb <- roc(y_test, pred_test_xgb, quiet = TRUE)

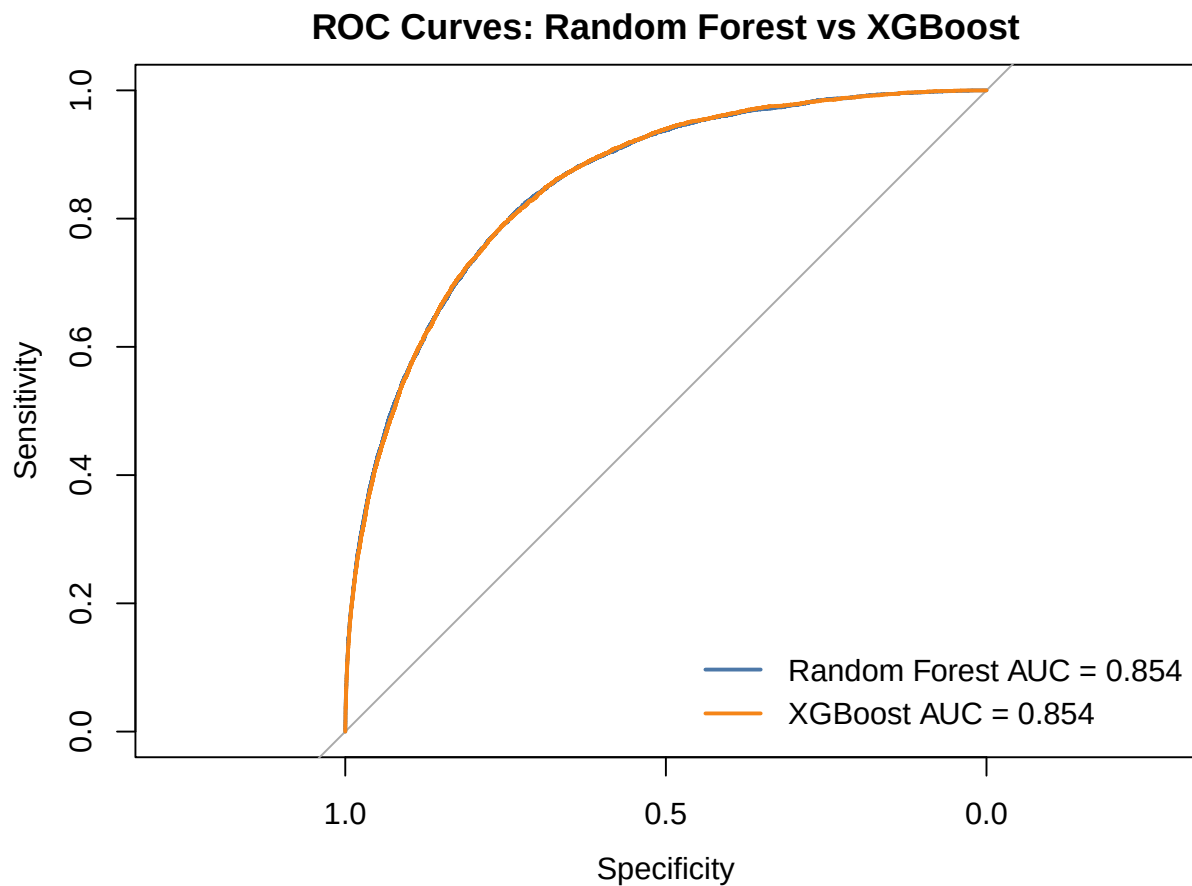
plot(
  roc_rf,
  col = "#4C78A8",
  lwd = 2,
  main = "ROC Curves: Random Forest vs XGBoost"
)
plot(
  roc_xgb,
  col = "#F58518",
  lwd = 2,
  add = TRUE
)

legend(
```

```

"bottomright",
legend = c(
  paste0("Random Forest AUC = ", round(as.numeric(auc(roc_rf)), 3)),
  paste0("XGBoost AUC = ", round(as.numeric(auc(roc_xgb)), 3))
),
col = c("#4C78A8", "#F58518"),
lwd = 2,
bty = "n"
)

```



Final model comparison

```

model_comparison <- rbind(
  evaluate_model(

```

```

    "Logistic (Weighted)",
    y_train,
    pred_train_w_glm,
    y_test,
    pred_test_w_glm
),
evaluate_model(
    "Logistic (Unweighted)",
    y_train,
    pred_train_u_glm,
    y_test,
    pred_test_u_glm
),
evaluate_model(
    "LASSO (Weighted)",
    y_train,
    pred_train_w_lasso,
    y_test,
    pred_test_w_lasso
),
evaluate_model(
    "LASSO (Unweighted)",
    y_train,
    pred_train_u_lasso,
    y_test,
    pred_test_u_lasso
),
evaluate_model(
    "GAM (Weighted)",
    y_train,
    pred_train_w_gam,
    y_test,
    pred_test_w_gam
),
evaluate_model(
    "GAM (Unweighted)",
    y_train,
    pred_train_u_gam,
    y_test,
    pred_test_u_gam
),
evaluate_model(

```

```

    "Random Forest",
    y_train,
    pred_train_rf,
    y_test,
    pred_test_rf
  ),
  evaluate_model(
    "XGBoost",
    y_train,
    pred_train_xgb,
    y_test,
    pred_test_xgb
  )
)

model_comparison <- model_comparison %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  arrange(desc(Test_AUC))

model_comparison

```

	Model	Test_AUC	Threshold	Accuracy	Sensitivity	Specificity
1	Random Forest	0.854	0.112	0.780	0.759	0.782
2	XGBoost	0.854	0.464	0.749	0.799	0.744
3	GAM (Unweighted)	0.850	0.078	0.746	0.799	0.740
4	GAM (Weighted)	0.849	0.087	0.758	0.785	0.756
5	Logistic (Weighted)	0.847	0.085	0.754	0.786	0.751
6	Logistic (Unweighted)	0.847	0.076	0.740	0.801	0.734
7	LASSO (Unweighted)	0.846	0.085	0.758	0.775	0.756
8	LASSO (Weighted)	0.845	0.085	0.741	0.794	0.735